How riskdep computes probabilities

Priors per layer, Monte Carlo, breakeven p*, outside views

Two questions, two probabilities

How likely is an incident?

- A prior per layer, per year
- From the hack record and the registry
- Drives the Monte Carlo of analyze

How likely must it be to leave?

- Breakeven p* of an incident
- Cost of leaving vs loss saved
- No prior needed: the reader judges

A position depends on layers

Settlement chain: Ethereum

Yield source (issuer): sUSDe, by Ethena

Pendle market: the PT/SY pool

Lending market: Morpho Blue, LLTV 91.5 %

Oracle: decides when liquidation happens

Exploit prior: from DefiLlama's hacks

$$p = \mathrm{clip}(r_{tier} \times m_{category} \times f_{age} ,\ 0.5\ \%,\ 15\ \%)$$

r_tier: hacks per protocol per year, same size band, last 5 years
m_category: the category's hack rate ÷ all categories, kept in 0.5–2
f_age: 2 if listed < 1 year, 1 if 1–3 years, 0.7 beyond

The terms, for the demo

Term	Value	Why
r_tier (> 100M USD)	2.66 %/yr	34 hacks / 256 protocols / 5 years
m: Lending (Morpho)	1.89	lending is hacked more than average
m: Basis Trading (Ethena)	0.50	kept at the floor of the clip
m: Yield (Pendle)	0.65	below average
Oracle misprice	2.68 %/yr	24 oracle exploits / 179 protocols / 5 y; ×2 if unverified

Market events: from the registry

A depeg is not an exploit: not in the hack record

registry.json sets p, severity and recovery, with a basis

- sUSDe: 12 %/yr, average depeg 8 %, 75 % recovered
- Basis: USDe near 0.65 on one CEX on 2025-10-10

Unknown issuer: 15 %/yr default, 60 % recovered

Severity: a Beta(a, b) draw, mean a / (a + b)

The demo's priors

Event	Per year	Severity mean	Over 30 days
sUSDe depeg	12.0 %	8.2 % (75 % back)	1.05 %
Morpho Blue exploit	5.03 %	9.1 %	0.42 %
Oracle misprice	2.68 %	10.0 %	0.22 %
Ethena exploit	1.33 %	28.6 %	0.11 %
Pendle market / SY exploit	1.21 %	13.8 %	0.10 %
Ethereum incident	1.00 %	0.5 %	0.08 %

From probabilities to losses

20,000 simulated months (simulate.py)

Each layer fails with its 30-day probability

A failure draws its severity from its Beta

propagate.py turns shocks into the position's loss

- True value vs oracle value: only the oracle liquidates

Only dependency: an issuer exploit brings a depeg

What the simulation gives (demo)

Measure	Value	Meaning
P(any loss)	1.91 %	months where something happened
P(liquidation)	1.07 %	months ending in a liquidation
Expected loss	6,265 USD	the average month
VaR 99 %	386,890 USD	exceeded in 1 % of months
CVaR 99 %	510,184 USD	worst 1 %: 98 % of the equity

The verdict uses them

EXIT if expected loss > carry + exit cost

REDUCE if carry < expected loss, or CVaR 99 % > half the equity, or the liquidation distance < the stress drop

STAY otherwise

Demo: REDUCE (6,265 > 5,697; CVaR 98 %; 5.2 % < 12.2 %)

Exit not worth it: 78,946 USD to leave

Breakeven p*: no prior needed

p* = cost of leaving now ÷ loss saved by leaving first

Cost: the yield given up on what can leave, over the horizon
Saving: loss if staying − loss if leaving first
Per year: 1 − (1 − p*)^(365 / horizon)
Leave only if you believe the incident is more likely than p*

Outside view: Nexus cover prices

How

- Implied p = price ÷ (loading × severity)
- Loading 2–5: capital cost and margin
- Gives a range, per year

How it is used

- Shown next to p*
- Never in a verdict
- Says what the cover pays on

Limits

Base rates, not an audit of each protocol

Layers independent, except issuer exploit → depeg

Market-event priors are judgements, with a basis

A 5-year hack window: DeFi changes fast

Carry assumes today's rates over the horizon